UNIVERSITÀ
DEGLI STUDI
DI MILANO

LA STATALE

MANIPOLAZIONE CON ROBOT MOBILI TRAMITE TECNOLOGIE OPEN SOURCE

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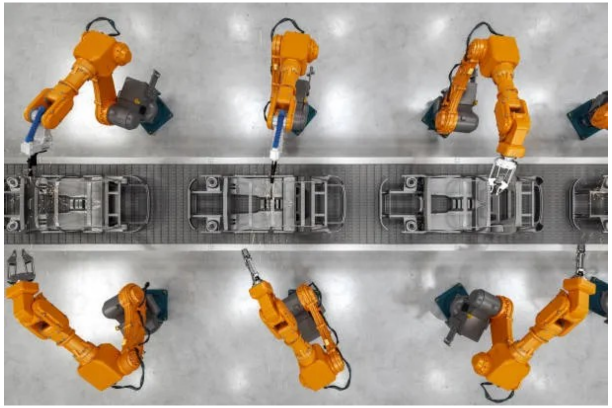
Robotica autonoma e Manipolazione

Ambiti di applicazione

Economia delle risorse



Perché le tecnologie Open Source?

Accessibilità

Community e risorse

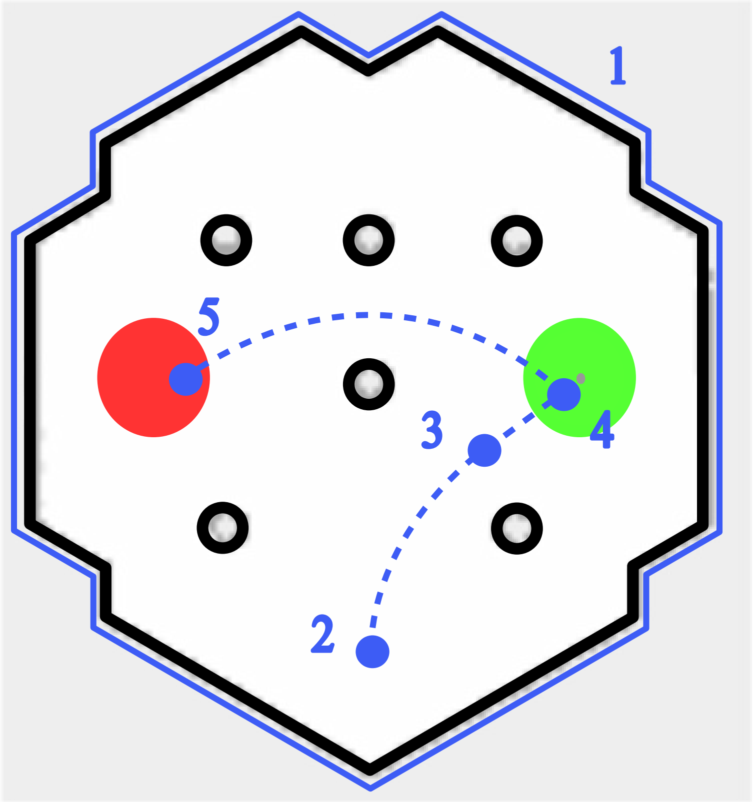
Vantaggi a lungo termine

Panoramica della tesi

Esecuzione di task complesse

Subtask di un pick and place

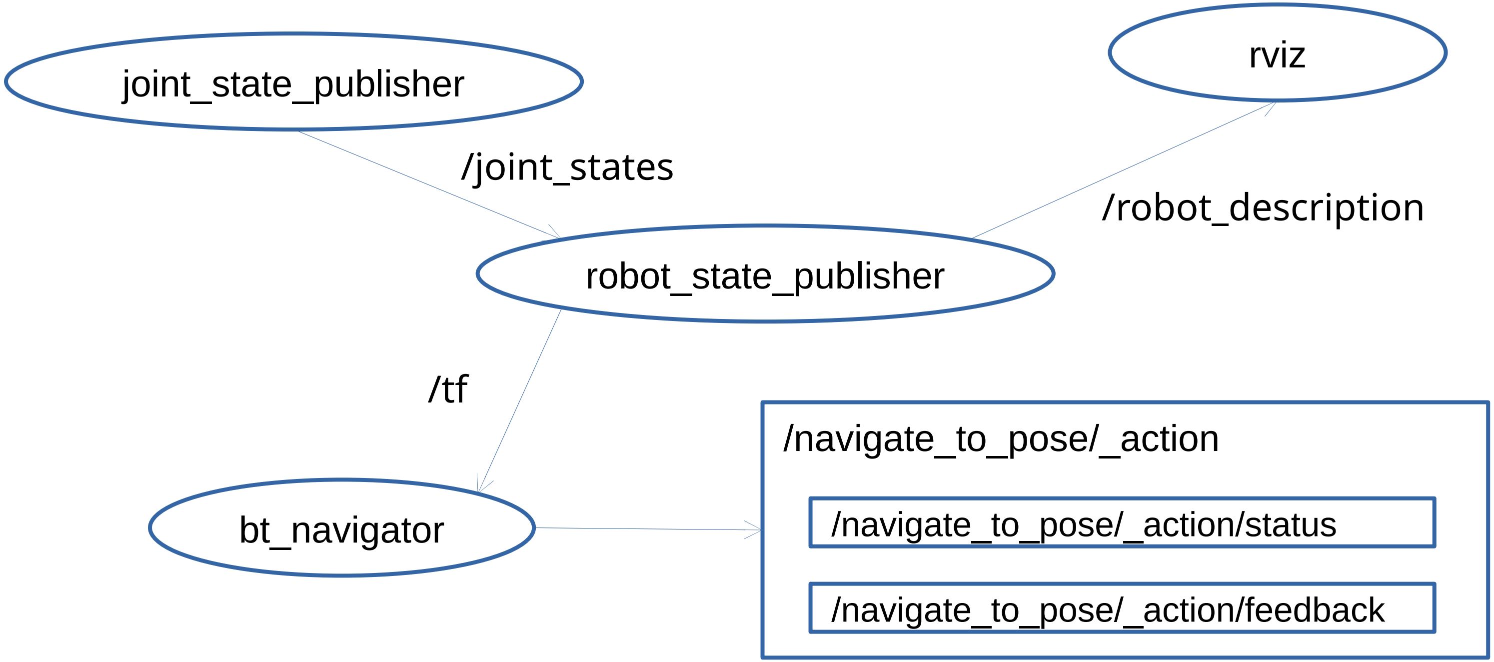
1. Generazione della mappa
2. Spawn
3. Navigazione autonoma
4. Object recognition e pick
5. Navigazione autonoma e place



Tecnologie di Base

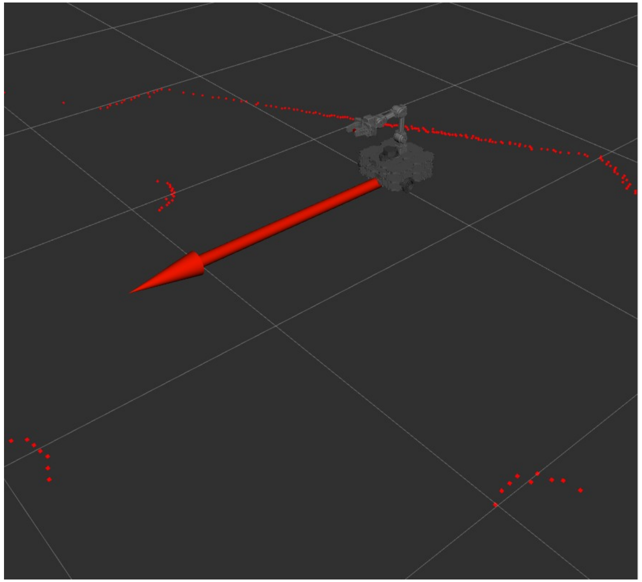
ROS 2

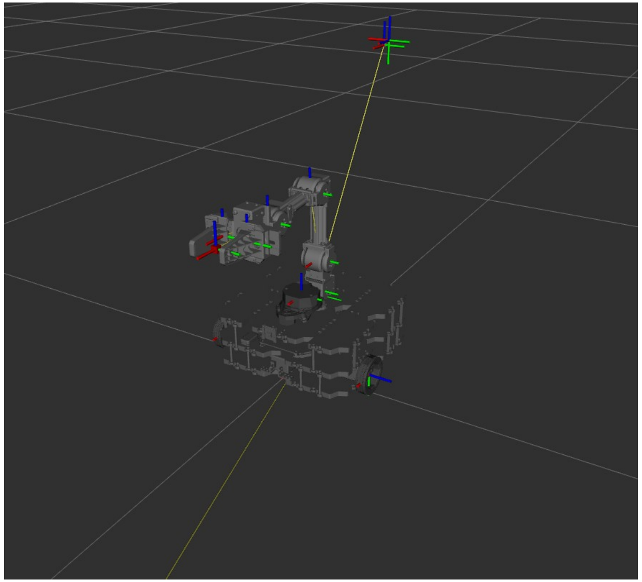
- Nodi
- Topic
- Service
- Action

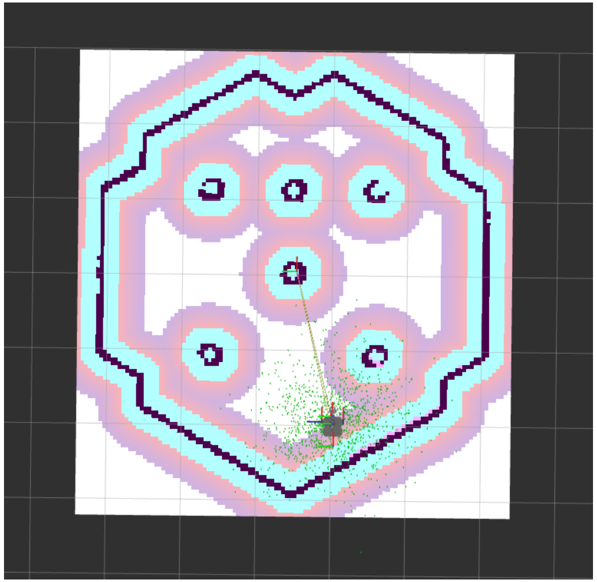


Tecnologie di Base

RVíz





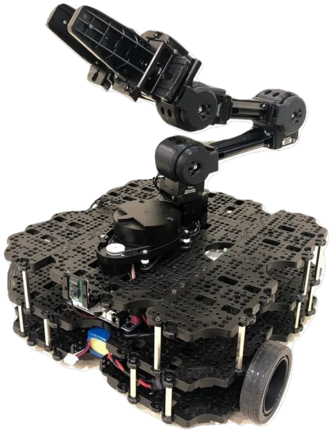


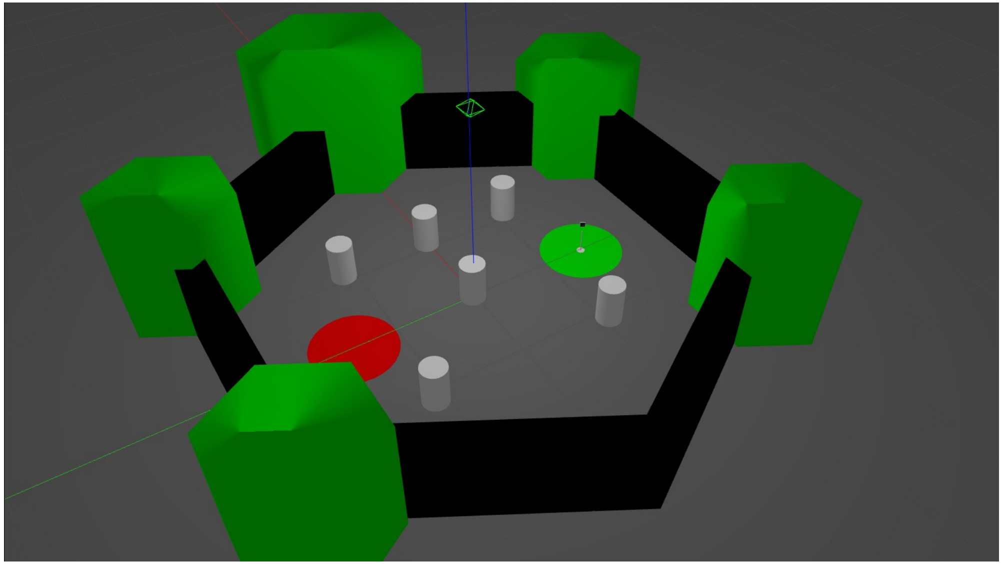
Simulazione

Gazebo

Formati

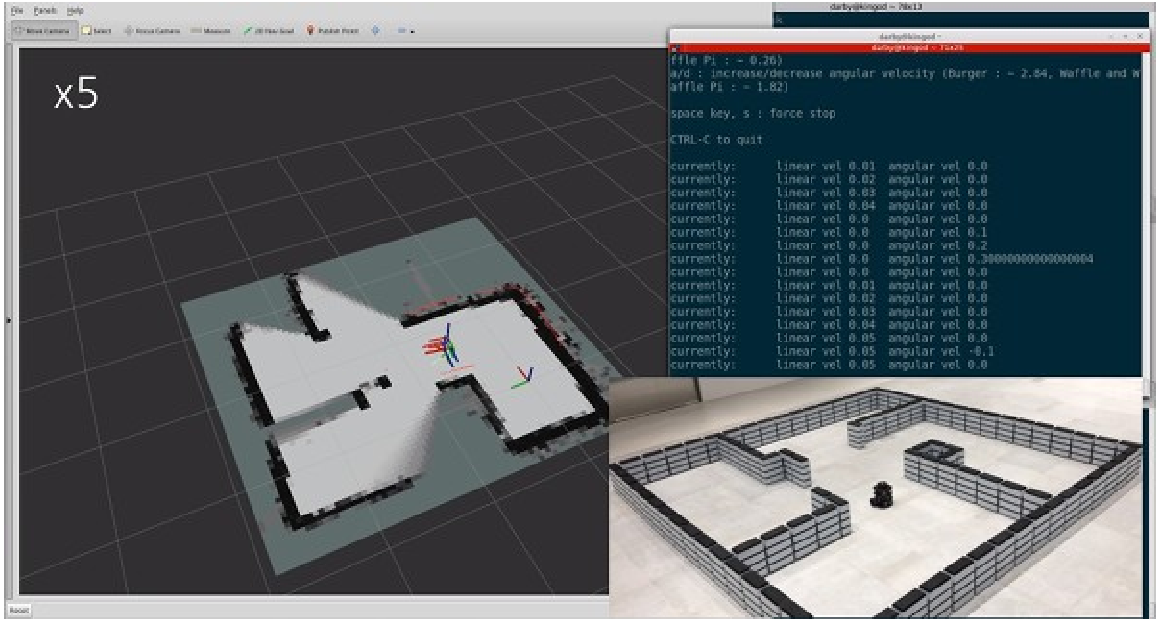
- URDF
- SDF
- Xacro





Per il progetto: robot, mondo, oggetti di scena

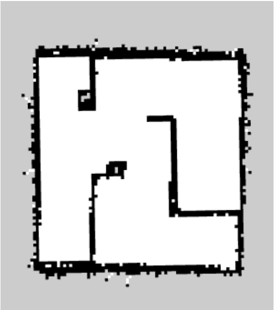
SLAM – Simultaneous Localization and Mapping



Localizzazione

Mapping: Occupancy Grid
Maps in formato PGM

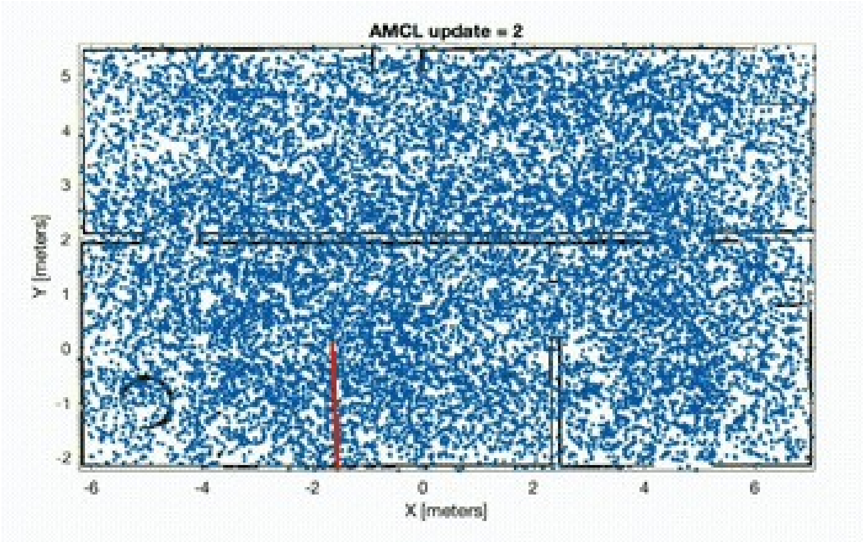
Slam Toolbox



Navigazione Autonoma

Navigation 2

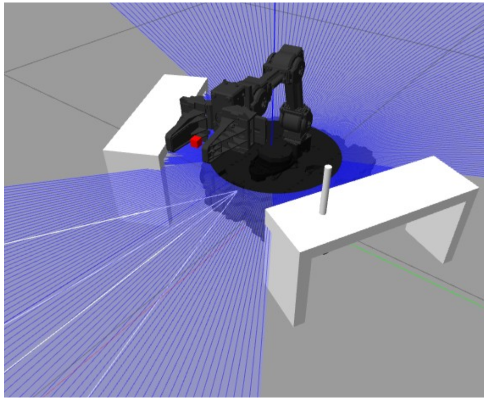
- Behavior Trees
- Recovery
- Layered Costmaps
- AMCL: Adaptive Monte Carlo Localization

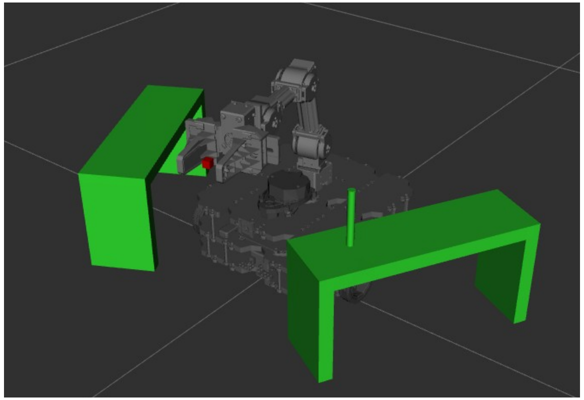


Motion Planning

MoveIt 2

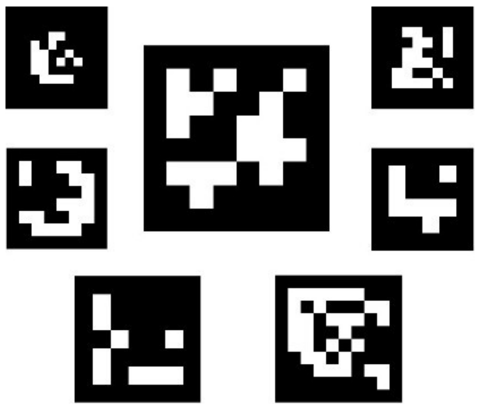
- Nodo move_group
- Planning scene
- Pick and Place

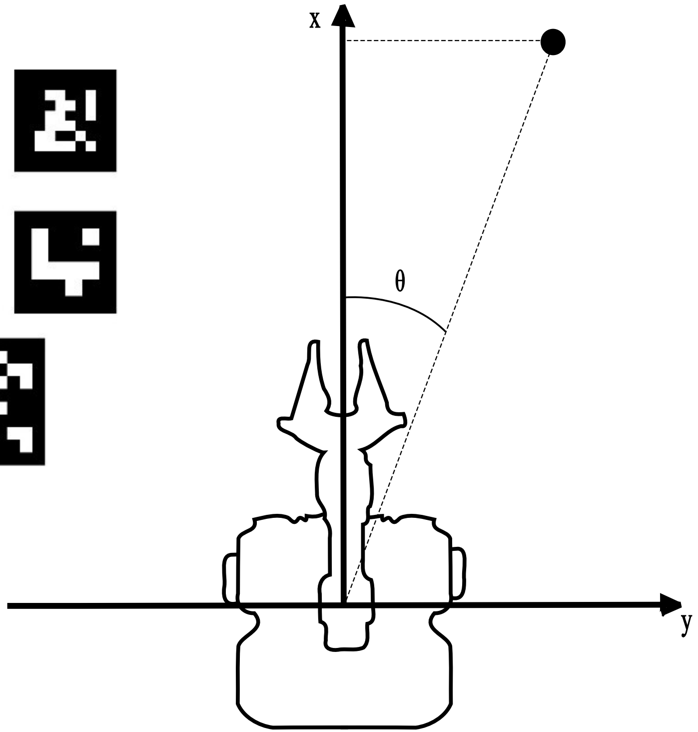




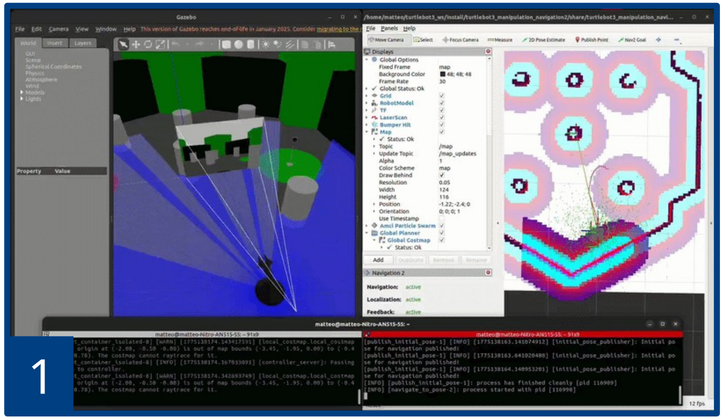
Sviluppo di Nodi Ad Hoc

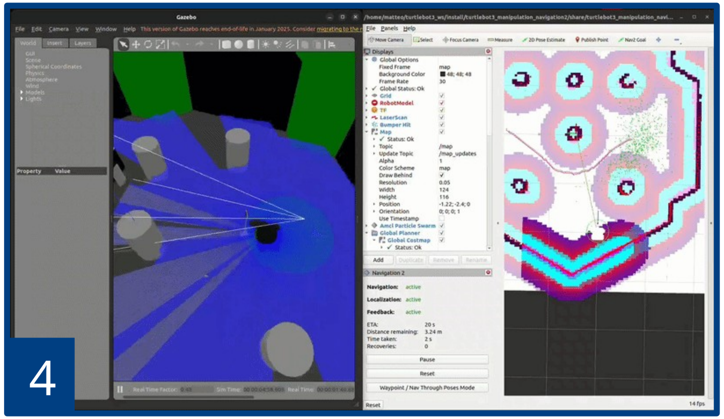
- *read_marker_pose* node
- *follow_marker* node
- *execute_pick* node
- *execute_place* node
- File di launch

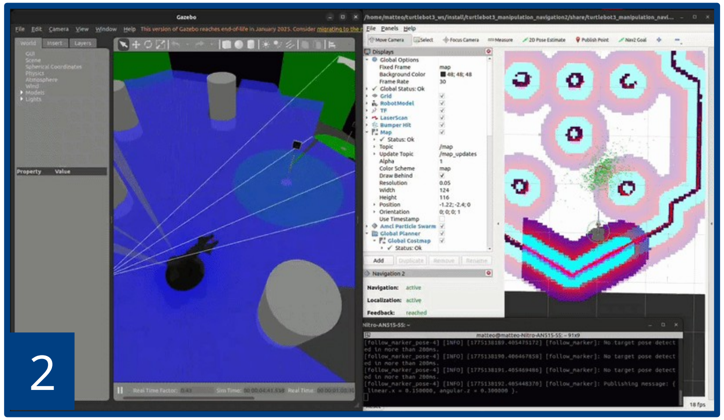


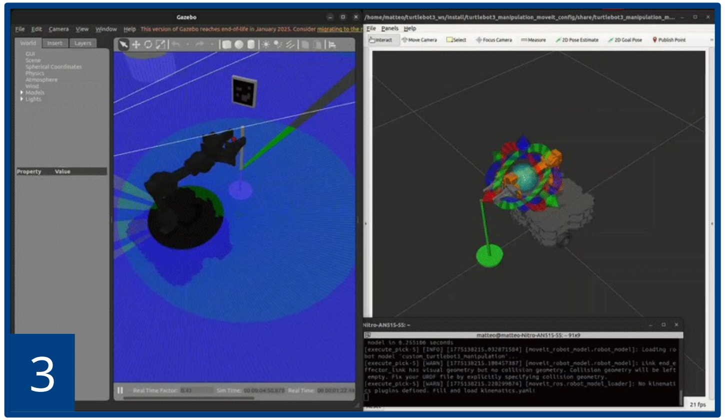


Risultati









Dimostrazione video: https://youtu.be/NwpsNIYv7PI

Codice sorgente: https://github.com/MatteoVignaga/Turtlebot3_Nav_PickAndPlace.git